

Systemic Vulnerabilities of Mechanical Differentials in Three-Wheeled Electric Architectures and the Transition to Electronic Torque Coordination Strategies

The drivetrain architectures of light commercial vehicles, particularly the ubiquitous three-wheeled platforms operating in dense urban environments, are undergoing a fundamental paradigm shift driven by the electrification of mobility. Historically, these delta-configuration tricycles—typified by platforms such as the Bajaj RE auto-rickshaw—have relied on centralized internal combustion engines mated to mechanical transaxles and conventional open differential units. The mechanical differential is an essential kinematic linkage designed to solve a fundamental geometric problem: allowing wheels mounted on a shared axle to rotate at disparate angular velocities, which is an absolute necessity for cornering without inducing severe tire scrub, driveline binding, or catastrophic loss of control. However, while the mechanical differential solves this basic kinematic requirement, it simultaneously introduces profound systemic limitations in vehicle efficiency, architectural packaging, dynamic stability, and long-term mechanical reliability.

With the rapid adoption of electric propulsion, the inherent mechanical constraints of conventional drivetrains are being rigorously evaluated against the advanced capabilities of Electronic Differential Systems (EDS). By decoupling the left and right drive wheels and utilizing independent electric motors—such as direct-drive in-wheel hub motors—engineers can emulate and vastly exceed the functionality of mechanical differentials via software-defined torque vectoring. Furthermore, as explicitly outlined in contemporary engineering challenges for next-generation auto-rickshaws, replacing a rigid rear axle or mechanical differential with independent rear hub motors introduces profound mechanical simplifications but removes the natural torque equalization provided by physical gearsets. This necessitates the development of sophisticated vehicle-level control concepts to manage stability, steering feel, and drivability. The exhaustive analysis provided herein scrutinizes the operational, mechanical, and dynamic issues inherent in mechanical differentials, addresses the unsprung mass debate, and establishes a conceptual framework for the required sensors, physical models, and torque coordination algorithms necessary to resolve these structural deficits through electronic architectures.

Fundamental Mechanics and the Traction Paradox of Mechanical Differentials

To thoroughly understand the systemic issues of the mechanical differential, it is necessary to deconstruct its physical constraints and operational logic. The most ubiquitous variant utilized in light commercial vehicles is the open differential. This system utilizes a planetary gearset housed within a rotating carrier. The carrier is driven by a ring gear, which is in turn driven by a pinion gear connected to the primary driveshaft originating from the prime mover. In a three-wheeled auto-rickshaw, these components are often consolidated into a single transaxle unit to simplify manufacturing, though this creates a dense, heavy assembly located at the rear of the chassis.

The Core Vulnerability: Torque Equalization and Spinout

The defining operational flaw of the open differential is its rigid adherence to torque equalization. By mechanical necessity, an open differential always splits torque equally (a strict 50/50 distribution) between the two driven wheels, regardless of their respective rotational speeds or the friction coefficient of the surface beneath them. Consequently, the absolute maximum amount of forward torque that can be applied to the vehicle's axle is strictly limited to twice the amount of torque that the wheel with the lowest coefficient of friction can sustain.

If one wheel encounters a low-friction surface—such as mud, ice, loose gravel, or an uneven road shoulder—and loses traction, the differential allows that wheel to spin freely. As the slipping wheel loses its ability to react against the road surface, the torque applied to it drops to near zero. Because the open differential must balance torque equally, the torque delivered to the opposing wheel (which may be resting on high-friction, dry pavement) also drops to near zero. This results in a complete loss of forward propulsion, a phenomenon commonly referred to as a "spinout". In the context of a heavily loaded three-wheeler operating on poorly maintained urban or semi-urban roads, this traction paradox severely limits the vehicle's operational reliability and gradeability.

The Compromises of Passive Limited-Slip Systems

To mitigate the traction paradox of the open differential, automotive engineering developed Limited-Slip Differentials (LSD) and fully locking differentials. Locking differentials mechanically bind the two axle half-shafts together, forcing them to rotate at identical angular velocities. While this entirely solves the traction issue by ensuring power reaches both wheels, it eliminates the differential action entirely. If engaged on high-friction surfaces, locking differentials induce severe drivetrain wind-up, aggressive tire scrubbing, massive steering understeer, and potentially catastrophic mechanical failure of the axle shafts.

Mechanical LSDs, including clutch-type units and helical gear (Torsen) differentials, rely on friction discs or complex gear binding geometries to transfer torque (often referred to as torque bias) away from the slipping wheel and redirect it to the gripping wheel. While theoretically superior to open differentials, these systems possess inherent flaws that make them unsuitable or highly compromised for low-power, high-efficiency electric platforms:

1. **Passive and Reactive Operation:** Mechanical LSDs are strictly passive devices. They require a slip event to physically occur before they can generate the internal frictional binding required to redistribute torque. This results in a delayed response time during

which vehicle stability is momentarily compromised.

2. **Torque Bias Ratio Limitations:** Torsen and similar gear-binding differentials multiply the torque of the slipping wheel by a fixed ratio (the Torque Bias Ratio) as plotted on a torque bias diagram. However, if a wheel is completely suspended in the air (possessing zero traction)—a common occurrence during the extreme lateral load transfer of a three-wheeler—multiplying zero torque by any ratio still results in zero torque transferred to the grounded wheel, rendering the vehicle immobile.
3. **Mechanical Complexity and Friction Wear:** Clutch-type LSDs suffer from continuous frictional wear on their internal clutch packs. This degrades their performance over time, alters their locking characteristics unpredictably, and requires stringent maintenance intervals and specialized friction-modifying lubricants to prevent chattering and premature failure.

Architectural Constraints: Mass, Packaging, and Center of Gravity

The physical dimensions, solid mass, and required operational clearances of a mechanical differential impose strict architectural constraints on vehicle design. These constraints conflict directly with the optimization goals of modern electric vehicles, which prioritize low mass, centralized battery placement, and modular chassis designs.

Volumetric Inefficiency and Ground Clearance Conflicts

In a standard rear-wheel-drive topology, the differential housing (colloquially known as the "pumpkin") is a large, bulky cast-iron or heavy-duty aluminum nodule situated at the exact center of the rear axle. This physical mass occupies a critical volume in the vehicle's undercarriage. In three-wheeled auto-rickshaws, designers are forced into a constant compromise between ground clearance and the vehicle's Center of Gravity (CG). Raising the axle and chassis to achieve adequate ground clearance over uneven terrain naturally elevates the vehicle's CG. Because a delta-configured three-wheeler (one front wheel, two rear wheels) possesses a triangular support polygon, a higher CG drastically increases the vehicle's propensity to roll over during lateral acceleration. Conversely, attempting to lower the floorpan to lower the CG and improve passenger ingress/egress is physically blocked by the presence of the differential housing, the transaxle assembly, and the solid axle tubes. The necessity of packaging these mechanical components forces the chassis floor upward, compromising both stability and cargo volume.

The Impact of Integrated Transaxles in Auto-Rickshaws

To understand the scale of this packaging issue, one must examine the specifications of standard commercial platforms. The Bajaj RE auto-rickshaw operates with a Gross Vehicle Weight (GVW) of approximately 672 kg, a kerb weight of 363 kg, and a wheelbase of 2000 mm. The rear axle and transaxle assembly is a heavy, integrated unit designed for durability rather than lightweight efficiency. The mechanical differential within this transaxle requires a

constant volume of lubricating oil, sealed bearings, and robust cast housings. When transitioning to electric variants, retaining this mechanical differential means retaining its mass, which directly subtracts from the allowable payload capacity or battery mass.

Architectural Feature	Mechanical Differential & Solid Axle	Independent Rear Hub Motors	Implications for Vehicle Design
Central Undercarriage Volume	Occupied by differential housing and transaxle	Completely unobstructed	Hub motors allow for a "low-floor" design, lowering the CG and increasing cargo space.
Component Mass	High (Cast iron housings, gearsets, fluid, shafts)	Relocated to wheels; overall drivetrain mass often reduced	Eliminating the differential offsets the mass of the hub motors, preserving the GVW limits.
Ground Clearance	Limited by the lowest point of the differential housing	Dictated only by the suspension arms and wheel diameter	Hub motors prevent high-centering on obstacles that would normally strike the differential housing.

Parasitic Power Loss and Mechanical Inefficiency

Beyond spatial and mass constraints, mechanical differentials are a significant source of continuous parasitic power loss. Power generated at the prime mover (whether an internal combustion engine or a central electric motor) must traverse multiple gear meshes, bearings, and fluid baths before finally reaching the tire contact patch. Within a typical vehicle, total drivetrain losses range from 10% to 25%, depending on the architecture (Front-Wheel Drive, Rear-Wheel Drive, or All-Wheel Drive).

The Thermodynamics of Drivetrain Loss

In a rear-wheel-drive configuration utilizing a longitudinal shaft and a 90-degree hypoid gear mesh at the differential, mechanical losses are particularly pronounced. Empirical energy flow analyses indicate that the differential and axle alone account for approximately 1.6% to 2.0% of direct power loss under optimal, steady-state conditions. However, this loss is not static; it fluctuates based on load, temperature, and speed. The inefficiency is partitioned into three primary mechanical domains:

1. **Gear Mesh Inefficiency:** Friction is generated continuously at the contact points of the ring and pinion gears, as well as the internal spider gears during cornering. Hypoid gears, which allow the pinion to be offset from the centerline of the ring gear, feature a sliding action along the gear teeth rather than a pure rolling action. This sliding

generates substantial heat and resistance, requiring energy to overcome.

2. **Viscous Fluid Churning:** To survive the extreme pressure of the sliding gear mesh, differentials are filled with high-viscosity gear oil (typically SAE 80W-90). As the ring gear rotates, it must continuously plunge through and churn this thick fluid bath. This viscous drag results in a direct loss of kinetic energy, particularly in cold environments where the fluid is highly viscous.
3. **Bearing Drag:** The differential carrier and the pinion shaft are supported by heavy-duty tapered roller bearings. These bearings require a specific mechanical preload to maintain gear alignment under torque. This preload introduces constant rolling resistance and friction.

In low-power light commercial vehicles, such as an e-rickshaw utilizing a 650W to 1500W electric motor, any parasitic loss represents a critical reduction in vehicle capabilities. A 15% drivetrain loss on a 1000W motor eliminates 150W of usable propulsion power. This manifests as reduced gradeability on steep inclines, sluggish acceleration under maximum payload, and a direct reduction in battery range per charge cycle.

Mechanical Failure Modes, Reliability, and Fleet Maintenance Overhead

The mechanical differential in a commercial three-wheeler is subjected to immense cyclical loading, shock loads from uneven terrain, and relentless thermal stress. The structural integration of the differential, gearbox, and axle into a single transaxle unit compounds these stresses and concentrates potential failure points.

Fatigue, Wear, and Fracture Mechanics

Metallurgical and structural failures within mechanical differentials are heavily documented in field maintenance reports and failure mode and effects analysis (FMEA) literature.

A prominent failure mode is torsional fatigue fracture occurring at the spline roots of the axle half-shafts. Driven by repeated acceleration and deceleration shock loads—common in stop-and-go urban traffic—micro-cracks initiate at machining marks or stress risers near the spline root. Over time, these cracks propagate transversely through the shaft due to the cyclic torsional loading until complete rupture occurs, instantly severing power transmission to the wheel.

Other severe mechanical failure modes include:

- **Gear Tooth Wear, Scoring, and Spalling:** Under high torque loads, or when lubrication breaks down due to overheating, the thin oil film separating the gear teeth is penetrated. This leads to metal-to-metal contact, resulting in adhesive wear, scoring (deep scratches), and pitting of the gear surfaces. As the surface hardens and chips (spalling), the debris circulates in the oil, accelerating the destruction of bearings and other gears until bending fatigue causes the gear teeth to shear off entirely.
- **Bearing Degradation:** The tapered roller bearings are highly susceptible to particulate contamination and improper preload. As bearings wear, they alter the critical backlash distance between the ring and pinion gears. Imminent bearing failure is diagnostically

identified by a high-pitched howling, whining, or humming noise emanating from the rear axle that varies with vehicle speed and throttle application.

- **Thermal Degradation and Seal Rupture:** Prolonged operation under heavy loads elevates the differential fluid temperature. This causes the oil to oxidize, thicken, and lose its extreme-pressure properties. Furthermore, excess heat degrades the elastomeric pinion and axle seals. A ruptured seal allows the gear oil to leak (observable as fluid pooling on the exterior housing or brake backing plates), resulting in fluid starvation, rapid overheating, and catastrophic internal seizure.

The Economic Burden in Commercial Fleet Operations

For high-utilization commercial fleets, the mechanical differential demands rigorous, recurring, and costly maintenance. Service protocols dictate routine fluid level checks, complete gear oil evacuations at specific mileage intervals (often every 30,000 to 50,000 miles), and periodic seal replacements. The downtime associated with rebuilding a differential—which requires skilled labor to set gear backlash, apply marking compound to check gear meshing patterns, and precise torque measurements to adjust bearing preloads—represents a severe economic penalty for fleet operators.

Failure Mode	Structural Mechanism	Diagnostic Symptoms	Operational Impact & Maintenance
Spline Torsional Fatigue	Cyclic shock loading propagating micro-cracks at machining stress risers.	Sudden loss of propulsion; one wheel spins freely without moving the vehicle.	Complete immobilization; requires disassembly of axle housing and half-shaft replacement.
Gear Tooth Scoring/Spalling	Lubrication boundary film breakdown; extreme pressure overload.	Clunking, whining, or howling noises during acceleration or deceleration.	Accelerated metal contamination; necessitates complete replacement of ring and pinion gearset.
Bearing Degradation	Abrasive wear from contaminated oil; improper mechanical preload.	Speed-dependent humming or whining noise; excessive play in the rear wheels.	Increased parasitic drag; altered gear backlash leading to rapid gear destruction.
Seal Rupture & Fluid Loss	Thermal degradation of elastomers; excessive shaft vibration.	Gear oil leaking onto the ground or brake assemblies.	Fluid starvation leading to rapid overheating, internal welding, and total differential lockup.

Dynamic Instability and Rollover Propensity in Delta

Configurations

The geometric architecture of a delta-configured three-wheeler (one steerable front wheel, two driven rear wheels) creates an inherently perilous dynamic stability profile. Understanding the physics of this instability is crucial for evaluating why a mechanical differential is a massive liability during extreme driving maneuvers.

The Physics of Tricycle Rollover

A three-wheeled vehicle relies on a triangular support polygon defined by the contact patches of the three tires. Because the engine, differential, payload, and passenger seating are primarily located over or slightly ahead of the rear axle, the vehicle's Center of Gravity (CG) is heavily rear-biased.

When the three-wheeler initiates a steering maneuver, lateral acceleration generates a centrifugal force vector acting horizontally through the CG. In a four-wheeled vehicle, the front track width provides a wide stance to resist this overturning moment. However, because the three-wheeler has a front track width of zero (a single wheel), the effective moment arm resisting the roll is a diagonal line drawn from the front tire to the outer rear tire. This drastically reduces the vehicle's roll stiffness.

Consequently, three-wheelers experience extreme Lateral Load Transfer Ratios (LTR) during cornering. The vehicle rapidly shifts weight off the inner rear wheel and onto the outer rear wheel and front wheel. When lateral acceleration exceeds a critical threshold—which can be as low as 0.3g to 0.4g for high-CG auto-rickshaws—the vertical load on the inner rear wheel drops to zero. The wheel lifts off the ground, and the vehicle enters an un-tripped rollover state. Furthermore, if the vehicle begins to slide laterally and the outer tires strike a pothole, curb, or soft soil, a tripped rollover occurs with violent suddenness.

The Inadequacy of Mechanical Differentials in Yaw Control

During these dynamic maneuvers, a mechanical differential is not merely useless; it actively exacerbates the danger. Because an open differential passively routes torque based on the path of least resistance, its behavior during a high-speed turn is detrimental. As the inner rear wheel unloads due to lateral load transfer, its traction limit plummets. The mechanical differential instantly senses this loss of resistance and shunts the entirety of the powertrain's torque to this lifting inner wheel.

This causes the inner wheel to spin wildly in the air, while simultaneously depriving the heavily loaded outer wheel of any driving force. The vehicle loses forward momentum mid-corner. Furthermore, a mechanical differential is incapable of proactively generating a stabilizing yaw moment. If the three-wheeler begins to oversteer (spin out) or understeer (plow straight), the mechanical differential has no mechanism to correct the vehicle's trajectory. It is entirely blind to the vehicle's dynamic state and is purely a slave to the physics of the tire contact patch.

The Electronic Differential Paradigm: Overcoming

Mechanical Deficits

Electronic Differential Systems (EDS) abandon passive physical linkages in favor of active, software-defined torque coordination. By decoupling the left and right wheels and utilizing independent motors—either through dual-motor e-axles or direct-drive in-wheel hub motors—an EDS can command specific, mathematically optimized torque values to each wheel instantaneously. This architecture directly addresses the mass, packaging, efficiency, and dynamic stability flaws of the mechanical system.

The Transition to In-Wheel Hub Motors and the Unsprung Mass Debate

As highlighted by the vehicle-level control concept constraints, replacing the mechanical differential with independent rear hub motors enables extreme mechanical simplification. It eliminates the transaxle, the driveshafts, the differential gears, and the heavy fluid housings. This frees up the entire undercarriage, allowing for an ultra-low floor design that drops the CG and maximizes passenger and cargo volume.

However, this transition shifts significant mass from the vehicle's chassis (sprung mass) directly into the wheels (unsprung mass). Unsprung mass is heavily scrutinized in vehicle dynamics engineering. A heavier wheel assembly has greater inertia; it is more difficult for the suspension dampers to push the heavy wheel down into depressions or absorb the upward shock of bumps. Theoretically, high unsprung mass causes the wheel to lose contact with the road surface more frequently, degrading both mechanical grip and passenger ride quality by transmitting harsh vibrations into the chassis.

Yet, empirical studies challenge the severity of this drawback in modern applications. A comprehensive dynamics study conducted by Lotus Engineering evaluated the impact of adding 36 kg of unsprung mass per corner via hub motors. The study utilized telemetry and subjective driver evaluations on both smooth tracks and degraded roads. The empirical conclusion was profound: differences in road surface roughness caused vastly larger variations in ride quality scores than the influence of the unsprung mass itself. With appropriate suspension retuning—specifically adjusting the damping rates for rebound and compression to manage the heavier wheel inertia—the degradation in ride quality perceived by the average driver is negligible. Crucially, the massive dynamic handling benefits unlocked by the independent torque control of hub motors—specifically active yaw control and torque vectoring—far outweigh the mechanical detriments of the increased unsprung mass.

Torque Vectoring vs. Brake-Based Emulation

When discussing electronic differentials, it is vital to distinguish between true torque vectoring and brake-based EDS emulation.

Many budget electric vehicles and modern internal combustion vehicles retain a mechanical open differential but use the Anti-lock Braking System (ABS) hardware to clamp the brake caliper on a slipping wheel. This artificially creates resistance at the slipping wheel, tricking the open differential into sending torque to the opposite wheel. While this is a cost-effective

software overlay, it is highly inefficient. It dissipates kinetic energy as heat, exacerbates brake pad wear, and reduces overall vehicle momentum during cornering.

In contrast, True Torque Vectoring (the focus of advanced EDS) utilizes independent dual motors to actively apply positive driving torque to the outer wheel and negative (regenerative) torque to the inner wheel. This generates a powerful, active yaw moment that physically rotates the vehicle into the turn, sharpening cornering response, neutralizing understeer, and maintaining vehicle speed without wasting energy to friction braking.

Conceptual Model for Vehicle-Level Torque Coordination

Addressing the specific control challenge of an auto-rickshaw equipped with independent rear hub motors requires defining a comprehensive conceptual model. Because the mechanical linkage is removed, the vehicle loses its natural torque equalization. If the left and right motors are sent identical torque commands while the vehicle is turning, the outer wheel will drag, and the inner wheel will spin, causing severe instability.

To achieve acceptable longitudinal performance and stable turning behavior under normal driving conditions without embedding complex hardware, the vehicle-level control concept must define the requisite sensors, control intent, and broad torque coordination algorithms.

Required Sensor Integration

To execute an electronic differential strategy, the vehicle's Electronic Control Unit (ECU) requires real-time telemetry regarding the driver's intent and the vehicle's dynamic state. The conceptual model necessitates a minimal but precise sensor suite:

1. **Steering Angle Sensor (SAS):** Located on the steering column/handlebar, this sensor provides the exact steering angle (δ_f). This is the primary input to determine the driver's intended turning radius and path.
2. **Wheel Speed Sensors (WSS):** High-resolution encoders located on each wheel to measure angular velocity. This data is critical for calculating longitudinal velocity, detecting individual wheel slip, and monitoring tire slip ratios.
3. **Inertial Measurement Unit (IMU):** Placed near the vehicle's center of gravity, the IMU measures the actual yaw rate (ω) and lateral acceleration (a_y). This feedback allows the ECU to compare the vehicle's actual behavior against the driver's intended path, enabling corrective torque vectoring.

First-Order Physics and Dynamic State Modeling

To coordinate torque broadly without delving into embedded PWM inverter complexities, the control intent utilizes simplified lumped mass physics and first-order vehicle motion equations.

The longitudinal motion of the vehicle depends on the sum of longitudinal tire forces:

$$m\dot{v}_x = F_{xfl} + F_{xfr} - F_{xrl} - F_{xrr} - F_{drag}$$

(where m is vehicle mass, $\dot{v}_x$ is longitudinal acceleration, and F_x represents tire forces).

Crucially, the yaw motion ($\dot{\psi}$), which dictates steering stability, is governed by the moment balance around the vertical axis (I_z). For a three-wheeler (ignoring the single front wheel's minimal lateral force contribution for simplicity), the yaw moment is heavily influenced by the difference in longitudinal forces generated by the independent rear wheels:

$$I_z\dot{\psi} \approx \frac{B}{2}(F_{xrr} - F_{xrl}) + M_{disturbance}$$

(where I_z is the moment of inertia, B is the rear track width, F_{xrr} and F_{xrl} are the longitudinal forces of the right and left rear tires respectively).

This first-order equation demonstrates the core premise of the electronic differential: by intentionally creating a differential in torque (ΔT) between the left and right hub motors, the controller directly manipulates the longitudinal tire forces ($F_{xrr} \neq F_{xrl}$), thereby generating a direct yaw moment to steer the vehicle or correct instability.

Advanced Control Strategies for Electronic Differentials

The execution of torque coordination relies on complex control algorithms. The literature extensively outlines two primary paradigms: Speed-based control and Torque-based control.

Speed-Based Control (Ackermann Kinematics)

In speed-based EDS, the controller utilizes a 2-Degree of Freedom (DOF) kinematic model to calculate the theoretical velocities required for the inner and outer wheels during a turn.

Utilizing the Ackermann-Jeantaud steering geometry, the ECU reads the steering angle (δ_f) and calculates the theoretical turning radius. It then computes the exact rotational difference needed so the inner and outer wheels traverse their respective arcs without slipping. The motor controllers are instructed to rigidly enforce these target speeds.

The Systemic Flaw: Speed-based control assumes a rigid adherence to theoretical geometry. In real-world dynamics, especially on uneven urban terrain, vertical suspension movement alters the dynamic rolling radius of the tire. Furthermore, at higher speeds, tires develop slip angles (the difference between where the wheel is pointing and where it is actually traveling). Forcing the hub motor to maintain a rigid, theoretically calculated speed under these dynamic load changes induces severe tire drag, conflicting slip ratios, and highly unpredictable vehicle instability.

Torque-Based Control and Optimum Slip Ratio

Torque-based EDS resolves the flaws of speed-based kinematics by utilizing dynamic force balancing. Rather than forcing a specific speed, the controller regulates the torque output based on the tire adhesion limits.

Within the stable range of tire attachment characteristics, road friction is a function of the longitudinal slip ratio (the difference between the rotational speed of the tire and the actual longitudinal speed of the vehicle). A torque-based EDS calculates an optimal slip ratio (typically 10-15% for maximum traction) and commands motor torque to keep the tire operating precisely within this optimal window. Because torque is commanded rather than speed, the wheel is allowed to rotate freely according to its specific vertical load and the road's friction limit. The tire naturally dictates its own optimal rotational speed, creating an adaptive steering effect that prevents wheel dragging over uneven terrain.

Hierarchical Control Architectures for Rollover Prevention

To ensure stable acceleration, braking, and turning while addressing the tricycle's rollover propensity, modern research proposes hierarchical, integrated control architectures. This methodology separates the control intent into distinct layers:

1. **Reference Model Layer:** Utilizes the steering angle and accelerator position to compute the driver's intended yaw rate and desired vehicle speed.
2. **Upper-Level Controller (Stability and Decoupling):** This layer acts as the dynamic safety net. It continuously monitors the IMU telemetry. If it detects that the actual yaw rate is deviating from the intended yaw rate (indicating understeer or oversteer), it calculates a corrective yaw moment. Crucially for three-wheelers, it utilizes Fuzzy Logic or Sliding Mode Control (SMC) to monitor the Lateral Load Transfer Ratio (LTR). If the LTR exceeds a safe threshold (indicating an imminent un-tripped rollover due to excessive lateral acceleration), the controller prioritizes rollover suppression over the driver's steering intent.
3. **Lower-Level Torque Allocator:** The allocator takes the requested corrective yaw moment and the total required driving torque and splits them into specific commands for the left and right hub motors.
 - *During Normal Turning:* It calculates the required differential torque (ΔT) and applies base torque (T_{base}) + ($\Delta T/2$) to the outer wheel, and base torque (T_{base}) - ($\Delta T/2$) to the inner wheel, ensuring smooth cornering without the drag of a locked axle.
 - *During Critical Instability:* If a rollover is predicted, the allocator applies severe negative (regenerative braking) torque to the outer front/rear wheels while reducing driving torque, rapidly decelerating the vehicle and generating a counter-yaw moment to drop the lifting inner wheel back to the pavement.

Implementation of the Control Loop System

To realize the torque-based control and hierarchical architecture described above, a closed-loop hardware and software implementation is required. The system continuously reads driver inputs and vehicle dynamic feedback to adjust motor torque dynamically, effectively bridging driver intent, vehicle physics, and motor actuation without requiring highly complex embedded inverter designs.

Overall System Block Diagram

The control loop operates continuously (typically at 100Hz to 500Hz) to read inputs, process vehicle dynamics, and command precise torque to individual motors.

INPUTS

(Steering, Throttle,
Brake)

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| VEHICLE CONTROL UNIT (VCU) |

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|| LAYER 1: Reference Model ||

|| (Calculates Ideal Dynamics based on inputs) ||

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| V |

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|| LAYER 2: Upper Controller ||

|| (Yaw Stability Control & Load Transfer Ratio Calculation) |<--+

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| V |

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|| LAYER 3: Torque Allocator ||

|| (Calculates Torque Split for Left/Right Wheels) ||

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| CAN Bus: T_left CAN Bus: T_right|
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V V
+-----+ +-----+
| MOTOR CONTROLLER | | MOTOR CONTROLLER |
| (Left) | | (Right) |
| | |
| <-- (Slip Control) | | (Slip Control) --> |
+-----+ +-----+ +-----+ +-----+
||
V V
+-----+ +-----+ +-----+ +-----+
| HUB MOTOR | | HUB MOTOR |
| (Left) | | (Right) |
+-----+ +-----+ +-----+ +-----+
||
+-----+
|
V
+=====+
| VEHICLE DYNAMICS |
| (Real-World Physics, Terrain & |
| Tire-Road Interaction) |
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|
V
+=====+
| SENSOR SUITE |
| (The Feedback) |
| |
| * IMU (Yaw Rate, Lateral Accel) |
| * Wheel Speed Sensors |
+=====+
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Feedback Loop

Detailed Block Implementation & Component Selection

Block A: Sensor Suite (The Feedback Loop)

To execute torque vectoring and rollover prevention, the system requires precise, real-time telemetry from the vehicle chassis and wheels.

- **Inertial Measurement Unit (IMU):** This is the most critical sensor for the Upper-Level Controller, measuring lateral acceleration (a_y) and yaw rate (ω). Automotive-grade units like the Bosch BMI088 are preferred as they are specifically designed to suppress harsh chassis vibrations found in high-vibration applications such as robotics and lightweight chassis.
- **Wheel Speed Sensors (WSS):** Required to calculate the true vehicle speed (from an un-driven wheel) and the individual slip ratios of the rear driven wheels. High-resolution magnetic sensor ICs, such as the Allegro A19352 utilizing Giant Magnetoresistance (GMR), provide extremely accurate speed and directional data directly from the wheel hubs.

Block B: Vehicle Control Unit (VCU)

The VCU serves as the central processing hub. To handle the complex matrix calculations for Sliding Mode Control (SMC) and Fuzzy Logic without latency, a robust 32-bit microcontroller, such as an STM32 series MCU running a real-time operating system (RTOS), is highly suitable for embedded vehicle stability applications. The VCU software is partitioned into three hierarchical layers:

- **Layer 1: Reference Model:** Translates driver steering angle (δ_f) and throttle inputs into an ideal mathematical path (desired yaw rate, ω_{des}) and a base torque request (T_{base}) based on a 2-Degree of Freedom (2-DOF) bicycle model.
- **Layer 2: Upper Controller (Stability & Decoupling):** Acts as the dynamic safety net. It compares the intended yaw rate with the actual yaw rate from the IMU feedback ($e = \omega_{des} - \omega_{actual}$). Utilizing SMC or Fuzzy Logic, it calculates a corrective yaw moment (ΔM_z). Simultaneously, it monitors lateral acceleration from the IMU to compute the Lateral Load Transfer Ratio (LTR). If LTR exceeds a critical threshold (e.g., > 0.8), the controller prioritizes rollover suppression, overriding steering intent to mandate a corrective yaw moment that drops the lifting chassis.
- **Layer 3: Torque Allocator:** Takes the total required driving torque and the corrective yaw moment and mathematically splits them into specific commands. For example, applying $T_{base}/2 + \Delta T/2$ to the outer wheel, and $T_{base}/2 - \Delta T/2$ to the inner wheel.

Block C: Communication Interface

The calculated torque commands must be transmitted to the motor controllers instantly and reliably. This is achieved using a Controller Area Network (CAN) bus. CAN is an automotive standard that utilizes differential signaling to ensure data integrity and prevent the heavy electrical noise generated by the EV powertrain from corrupting the commands.

Block D: Motor Controllers & Torque Execution

To execute Torque-Based Control and maintain the Optimum Slip Ratio without designing embedded inverters from scratch, the system utilizes commercial "smart" motor controllers capable of Field-Oriented Control (FOC), such as VESC hardware.

- **Torque Control Command:** Instead of commanding a rigid RPM (which causes dragging), the VCU sends a specific current target via a standard CAN command (e.g., CAN_PACKET_SET_CURRENT) over the bus to each individual controller.
- **Current Application:** The motor controller attempts to push exactly that current into the stator phases to generate the requested mechanical torque.
- **Slip Control Loop Integration:** The controller continuously monitors wheel speed via its internal Hall sensors or external WSS. If the wheel's rotational speed drastically exceeds the vehicle's forward speed (indicating a slip ratio above the optimal 10-15% threshold), a local anti-slip algorithm inside the motor controller instantly overrides the CAN bus torque request. It chops the electrical current just enough to drop the slip ratio back into the optimal window, ensuring the tire freely adapts to the terrain's friction limits without mechanical binding.

Economic Analysis, Total Cost of Ownership, and Efficiency Gains

The transition from a mechanical transaxle to a dual-hub motor electronic differential architecture involves a complex economic and efficiency calculus. While the upfront procurement cost of dual in-wheel motors and sophisticated dual motor controllers exceeds the cost of a single central motor mated to a rudimentary cast-iron transaxle, the Total Cost of Ownership (TCO) heavily favors the electronic architecture for commercial fleets.

Motor Efficiency Maps and Operational Domains

When evaluating efficiency, engineers must compare central mid-drive motors (which retain the mechanical differential) against dual hub motors. Mid-drive systems benefit from mechanical gear reduction. By utilizing multiple gear ratios, a smaller central motor can operate at high RPMs—where its electromagnetic efficiency map peaks—while delivering multiplied, low-speed torque for steep hill climbing.

Direct-drive hub motors do not utilize gearing. Consequently, their efficiency drops heavily during steep, low-speed climbs because the motor is forced to operate outside of its optimal thermal and rotational envelope, generating excess heat. However, in flat urban environments and stop-and-go traffic—the primary operational domain of a commercial auto-rickshaw—dual hub motors exhibit superior overall system efficiency. By completely eliminating the mechanical differential and drive shafts, the drivetrain avoids the constant 10-15% parasitic power drain caused by mechanical friction, bearing drag, and viscous fluid churning. Empirical coast-down and range tests comparing geared hub motors to direct-drive systems show that the elimination of the centralized mechanical drivetrain unequivocally improves battery utilization in urban scenarios, translating to extended range per charge cycle.

Maintenance Eradication and Fleet Uptime

The most compelling economic justification for electronic differentials is the total eradication of differential maintenance. An internal combustion engine vehicle contains over 2,000 moving parts, with the transmission and differential contributing significantly to mechanical wear. An electric vehicle utilizing dual hub motors reduces the drivetrain's moving parts to less than 25—primarily the motor rotors and sealed wheel bearings.

This architectural simplification yields profound economic benefits:

- **Fluid and Labor Savings:** There is no heavy hypoid gear oil to inspect or replace, completely eliminating the labor and material costs associated with 30,000-mile fluid service intervals.
- **Component Lifespan:** The eradication of splined half-shafts, spider gears, and friction clutch packs entirely removes their associated failure modes (torsional fatigue, gear spalling, bearing whine) from the vehicle's lifecycle. The vehicle will never experience downtime due to a shattered differential carrier.
- **Tire Conservation:** A poorly functioning mechanical differential, or an improperly specified limited-slip unit, causes severe tire scrubbing during tight cornering. By utilizing precision torque-based electronic control, an EDS ensures that wheels slip only within their optimal slip ratio, drastically extending the lifespan of the commercial vehicle's tires, which represents a massive ongoing operational expense.

Ultimately, the elimination of mechanical downtime, fluid maintenance, component replacement, and extended tire life creates a highly favorable break-even point for fleet operators, validating the transition from mechanical linkages to electronic torque coordination.